

C++ Implementation of Digital FIR Filters Using Blackman Window

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This utility program can be a handy tool for anyone working in the field of signal processing with interest in implementation of the project through C++ programming. Although this work can be easily implemented in Matlab using simple inbuilt functions, the real challenge of designing and understanding the concept of filter design will be missed out. In order to understand the depth of designing an FIR filter using window functions, coding in C++ is a must. Described here is a C++ implementation of finite impulse response (FIR) filters using Blackman window method.

FIR filters

The term filter is often used to denote systems that process the input in a specified way. In this context, filtering describes a signal-processing operation that allows signal enhancement, noise reduction or increased signal-to-noise ratio. Systems for the processing of discrete time signals are also called digital filters. Depending on the requirement and application, the analysis of a digital filter may be carried out in time domain, z-domain or frequency domain.

A common application of digital filters is to modify the frequency response in some specified way. An ideal low-pass filter is one of the common examples of digital filters which passes the frequencies up to a specified value and blocks other frequencies. One way to approximate the ideal low-pass response is to do truncation of the impulse response by multiplication with the window functions. Mathematically, it can be expressed as follows

$$h(n) = \begin{cases} h_d(n)w(n) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases} \quad \text{----- eqn(1)}$$

where $h_d(n)$ is an infinite-duration impulse response and $w(n)$ is the window function. There are various windows available in signal processing literature. However, in this article we specifically discuss Blackman window. In the case of Blackman window, $w(n)$ is given as:

$$w(n) = 0.42 - 0.5 \cos\left(\frac{2n\pi}{N-1}\right) + 0.08 \cos\left(\frac{4n\pi}{N-1}\right) \\ = 0 \quad \text{where } n=0, 1, 2, \dots, N-1 \quad \text{otherwise} \quad \text{----- eqn(2)}$$

Plot of eqn(2) can be seen graphically as shown in Fig. 1.

Software design

Here, we have developed C++ based software for designing FIR filters using Blackman window to get frequency response. The resultant software has the following salient features:

1. Welcome screen of Blackman window design

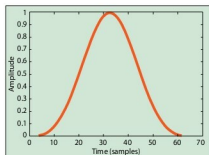


Fig. 1: Blackman window

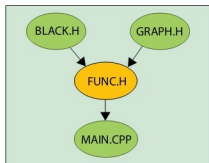


Fig. 2: Software flow



2. It prompts you to enter filter's type
3. It prompts you to enter window length (N) from 1 to 200
4. It prompts you to enter the cutoff frequency. The software is designed to accept the values in radians between 0.2 and 2.6
5. The program outputs normalised filter response. This is a filter response which is designed to pass through a cutoff frequency of 1 radian
6. The program outputs actual filter response. This is a filter response which is designed to pass through a cutoff frequency specified by the user

Software structure

The software is designed to follow the structure as shown in Fig. 2, where the details of each file are explained as given below.

1. **MAIN.CPP.** This file contains the menu-driven program for user inputs
2. **FUNC.H.** It is a header file containing code related to Blackman window coefficients calculation
3. **BLACK.H.** It is a header file containing code for supporting C++ graphics
4. **GRAPH.H.** It is a header file again to support graphic content in C++

Output responses of the filter

Now, the results obtained after running the software is presented for Blackman window for a low-pass FIR filter for different window length N to